

Keshav Rajput

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EDUCATION

New York University, Tandon School of Engineering

MS, Computer Science, May 2026. GPA 3.67/4.0

IIIT Delhi

B.Tech, Computer Science and Applied Mathematics; minor in Economics, 2020 - 2024. CGPA 8.19/10

Coursework: Applied Cryptography, Post-Quantum Cryptography, Computer Networks, Operating Systems, Distributed Systems, Design and Analysis of Algorithms II

PROJECTS

mlds-auth | C11, liboqs, libsodium, SQLite, CMake, libFuzzer github.com/happyc0der/mlds-auth, 2026

- Post-quantum mutual authentication and session protocol in C11: ML-DSA-65 signatures over hybrid X25519 + ML-KEM-768.
- 0.45 ms median handshake; ChaCha20-Poly1305 record layer at 736 MiB/s.
- 11 libFuzzer targets, 169 mutation tests, ASan/UBSan CI on Linux and macOS; login daemon with device enrolment and revocation.

One-time-pad allocation (team of 3) | Python, pytest github.com/happyc0der/cryptography-project-1, 2025

- Wait-free one-time-pad allocation protocol for m parties, with proofs that no pad is ever reused.
- Closed-form waste bound independent of n : 36 wasted pads against 4,000 for static partitioning ($m=5$, $n=5000$).
- Found two bugs in the assigned reference protocol; 310 tests, and CI re-checks the published results.

Hawk signature study | Python, NumPy, Flask github.com/happyc0der/hawk-signature-study, 2026

- Report and talk on Hawk, the lattice signature scheme broken and withdrawn from NIST standardisation in July 2026.
- Working keygen/sign/verify demo; six fixes to the reference Python code, one making verification about 300x faster.

Custodian | TypeScript, MCP, OAuth 2.1, Preact, AWS CDK, DynamoDB github.com/happyc0der/custodian, 2026

- MCP server and Alexa+ add-on that checks a household inventory against CPSC, NHTSA and FDA recall feeds. Built for Amazon Build, Ship, Shape 2026.
- Wrote the OAuth 2.1 authorization server: PKCE S256, rotating refresh tokens.
- TypeScript monorepo on AWS CDK (App Runner, DynamoDB); 10 tools, 78 tests, tool calls inside a 500 ms budget.

COMPETITIONS

- **Amazon Build, Ship, Shape 2026 (Alexa+ track)**, 2026. Shipped Custodian, an MCP recall-monitoring add-on.

SKILLS

Languages: Python, C, C++, Kotlin, TypeScript, JavaScript, SQL, Java, R, Bash

Security: libsodium, liboqs, ML-DSA, ML-KEM, X25519, ChaCha20-Poly1305, OAuth 2.1, libFuzzer, ASan/UBSan, threat modeling

Systems: CMake, gRPC, SQLite, WebAssembly, Android (Jetpack Compose, Room), Node.js, REST APIs, MCP

Tools: Git, GitHub Actions, Docker, AWS (CDK, App Runner, DynamoDB), Linux, LaTeX

ACTIVITIES

- Treasurer, OWASP Student Chapter, IIIT Delhi
- Attended the first International Quantum Communication Conclave, New Delhi, 2023